Computational Physics 2 Classes No.4 - Report

Imaginary Time Method – 2D Anisotropic Harmonic Oscillator

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1 Introduction

The aim of this report is to study a two-dimensional anisotropic harmonic oscillator using the imaginary time iteration method. The system is described by a harmonic potential with different frequencies in the two spatial directions. Numerical energy eigenstates are obtained on a finite two-dimensional grid and compared with the exact analytical energy values. The convergence of the method is analyzed for different values of the iteration parameter α, and the lowest excited states are computed using orthogonalization.

2 Results

The numerical calculations were performed for a two-dimensional anisotropic harmonic oscillator with

$$m = 1, \qquad \hbar = 1, \qquad \omega_x = 1, \qquad \omega_y = 2.$$

The computational domain was chosen as a square box

$$x, y \in [-2, 2],$$

with zero boundary conditions imposed at the grid boundaries. The grid size was

$$N = 200,$$

therefore

$$\Delta x = \Delta y = \frac{4}{N-1}.$$

The critical imaginary-time step was calculated as

$$\alpha_c = \frac{\Delta x^2}{2}.$$

For most calculations the stable value

$$\alpha = 0.9\alpha_c$$

was used.

The exact energy eigenvalues are

$$E_{n_x,n_y} = \omega_x \left(n_x + \frac{1}{2}\right) + \omega_y \left(n_y + \frac{1}{2}\right),$$

which for $\omega_x = 1$ and $\omega_y = 2$ gives

$$E_{n_x,n_y} = \left(n_x + \frac{1}{2}\right) + 2\left(n_y + \frac{1}{2}\right).$$

Thus, the first four exact energies are

$$E_{0,0} = 1.5, \qquad E_{1,0} = 2.5, \qquad E_{2,0} = 3.5, \qquad E_{0,1} = 3.5.$$

2.1 Ground-state convergence

The first part of the calculation was devoted to the convergence of the ground-state energy. Several values of the imaginary-time parameter were tested:

$$\alpha = 0.1\alpha_c, \quad 0.3\alpha_c, \quad 0.5\alpha_c, \quad 0.7\alpha_c, \quad 0.9\alpha_c, \quad 0.99\alpha_c, \quad 1.05\alpha_c.$$

For each value of α, the wavefunction was initialized randomly inside the computational box and normalized after every iteration.

Figure 1 shows the convergence of the energy expectation value for different values of α. The energy decreases during the imaginary-time evolution and approaches the exact ground-state value

$$E_{0,0} = 1.5.$$

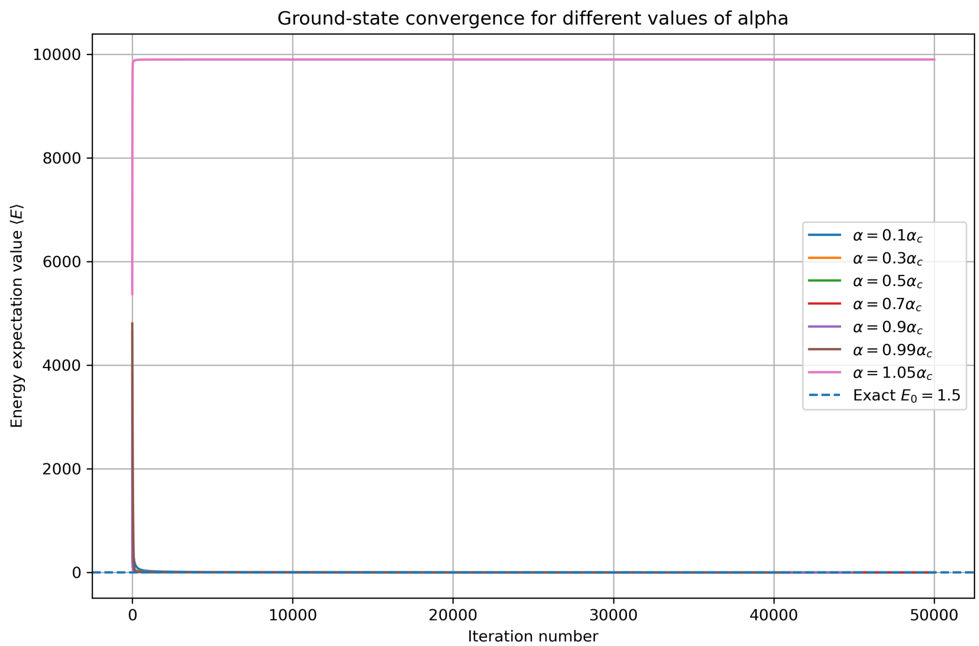

Figure 1: Ground-state energy convergence for different values of the imaginary-time parameter α.

A zoomed view near the final energy is shown in Figure 2. This plot makes it easier to compare the final converged energies with the exact analytical value.

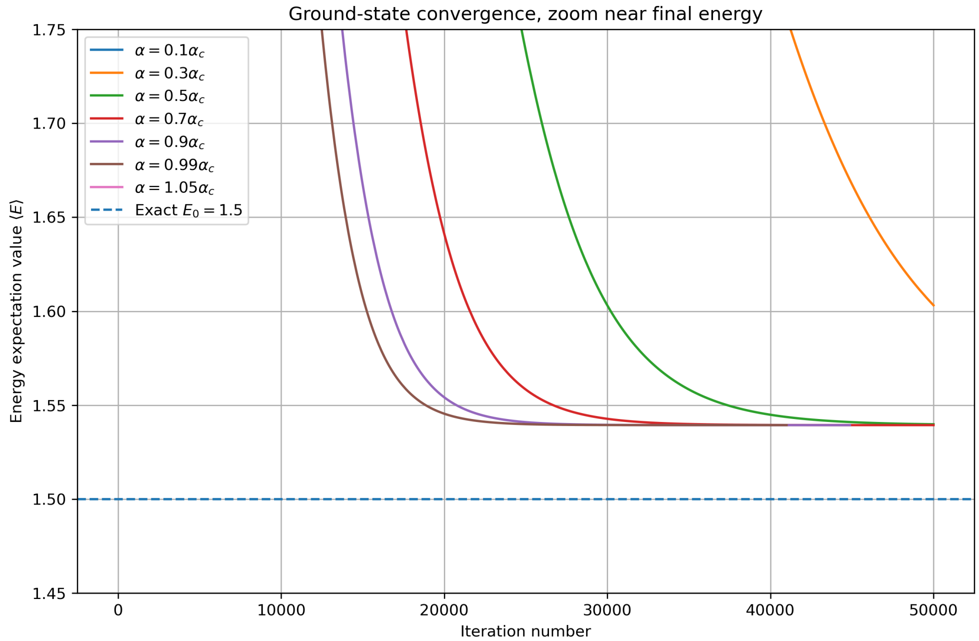

Figure 2: Zoomed view of the ground-state energy convergence near the exact value $E_{0,0} = 1.5$.

The error relative to the exact ground-state energy is shown in Figure 3. The logarithmic scale shows that the convergence is faster for larger stable values of α.

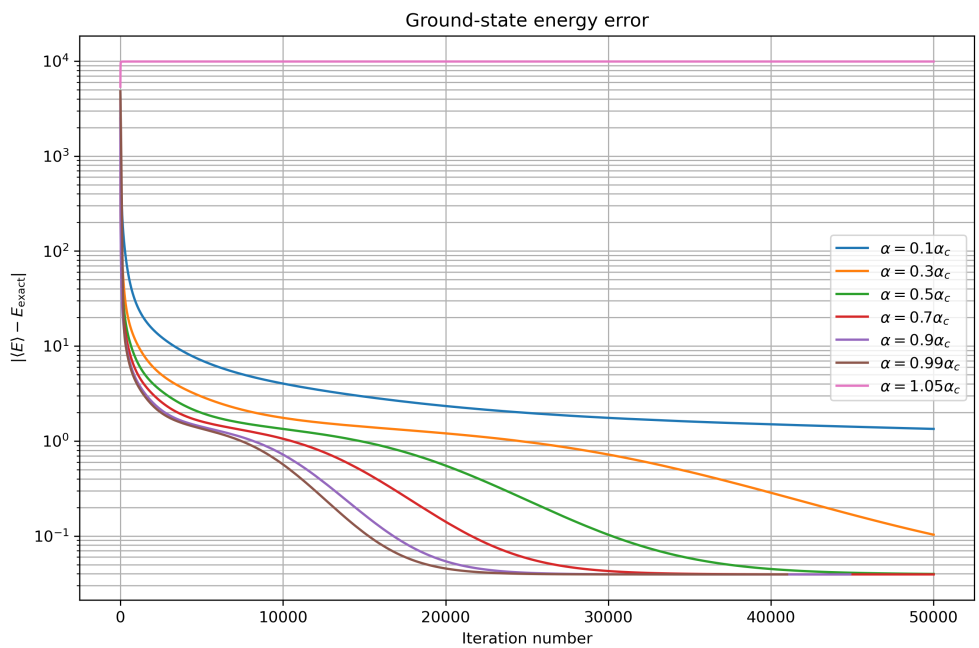

Figure 3: Absolute error of the ground-state energy relative to the exact value.

The results show that small values of α lead to slow but stable convergence. Increasing α improves the convergence speed. The best stable convergence was obtained for values close to the critical value, especially for

$$\alpha = 0.9\alpha_c$$

and

$$\alpha = 0.99\alpha_c.$$

However, when α is chosen above the critical value, the iteration may become unstable. Therefore, the value

$$\alpha = 0.9\alpha_c$$

was used as a safe choice for the excited-state calculations.

2.2 Excited states

The excited states were obtained by combining imaginary-time iteration with orthogonalization. After each iteration, the current wavefunction was orthogonalized with respect to all previously obtained states:

$$c_k = \sum_{i,j} \Psi_k(i,j)\Psi(i,j)\Delta x \Delta y,$$

$$\Psi(i,j) := \Psi(i,j) - \sum_k c_k \Psi_k(i,j).$$

The wavefunction was then normalized again.

Four states were computed in total. Their expected exact energies are

$$E_1 = 1.5, \qquad E_2 = 2.5, \qquad E_3 = 3.5, \qquad E_4 = 3.5.$$

The convergence of the four lowest states is shown in Figure 4. The first state converges to the ground-state energy, the second state converges to the first excited-state energy, and the third and fourth states converge to the degenerate level with energy close to 3.5.

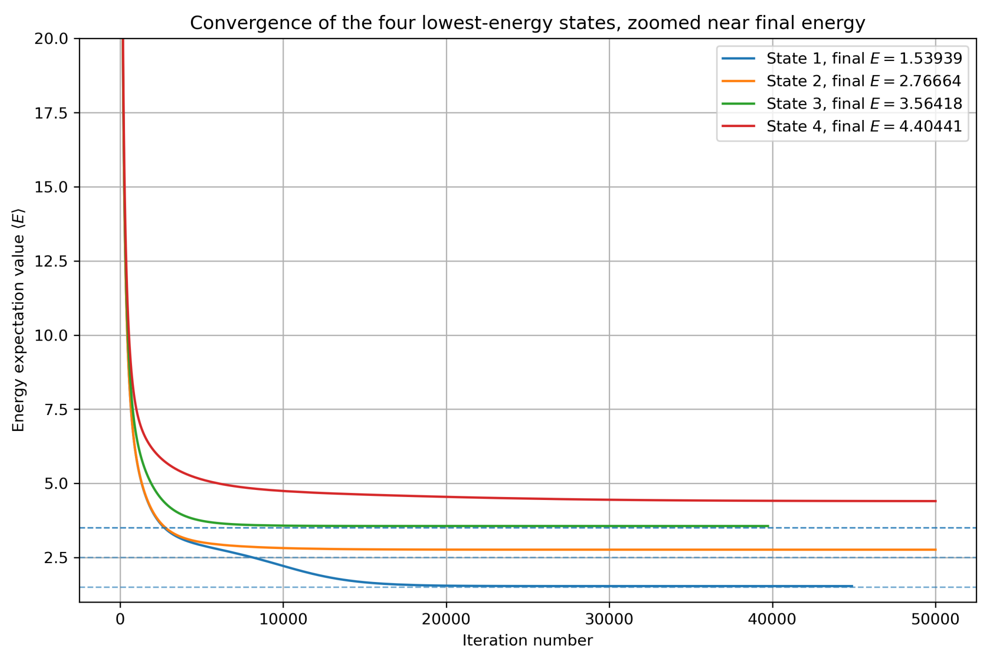

Figure 4: Energy convergence of the four lowest-energy states obtained with orthogonalization.

Separate convergence plots for each state are shown in Figures 5–8. These plots confirm that each numerical state approaches the corresponding analytical energy level.

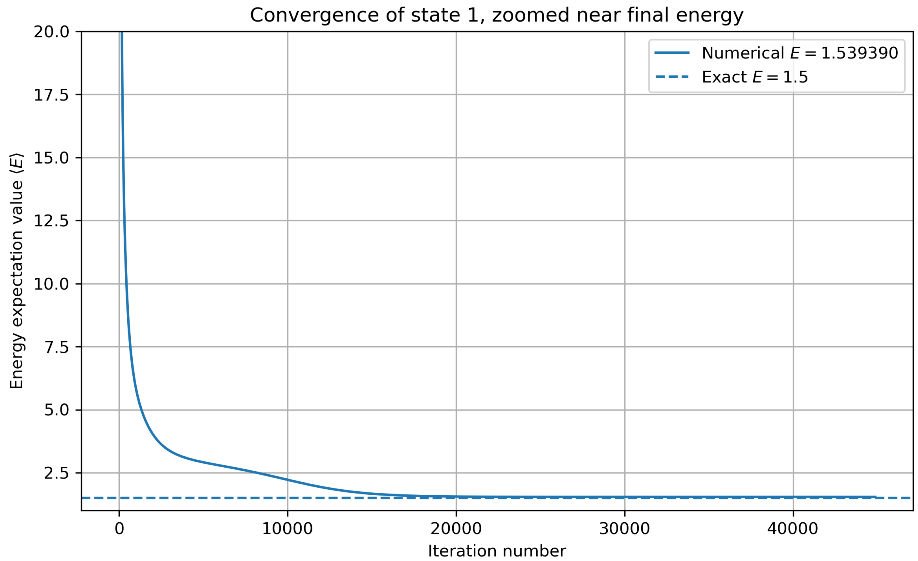

Figure 5: Energy convergence of the first state.

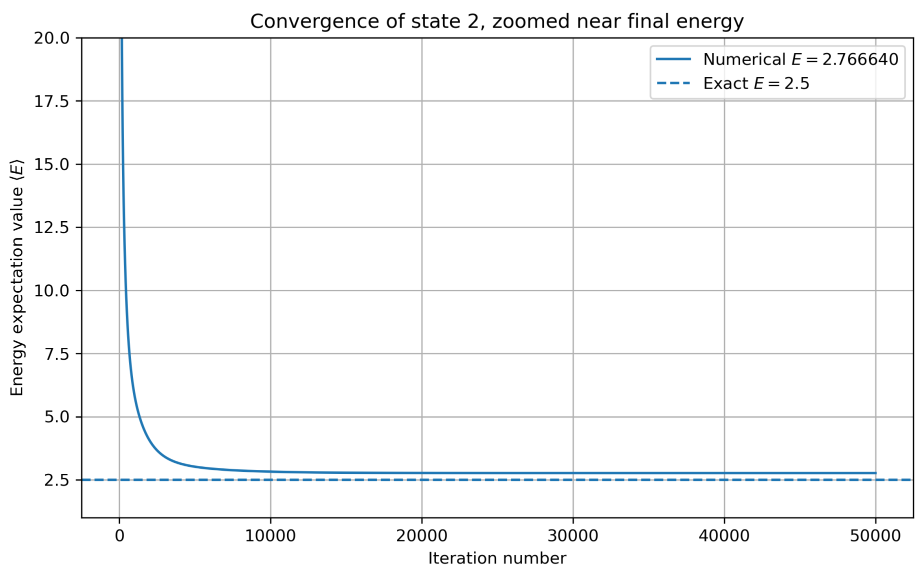

Figure 6: Energy convergence of the second state.

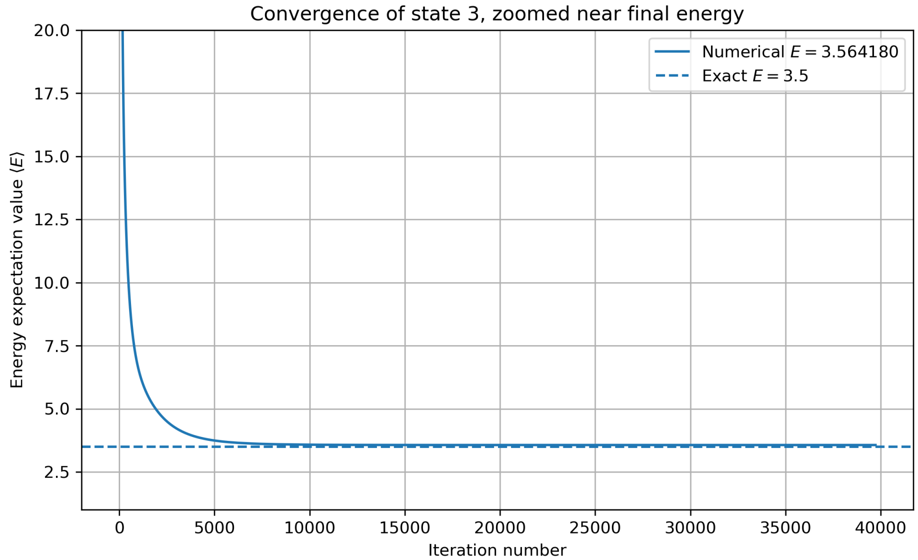

Figure 7: Energy convergence of the third state.

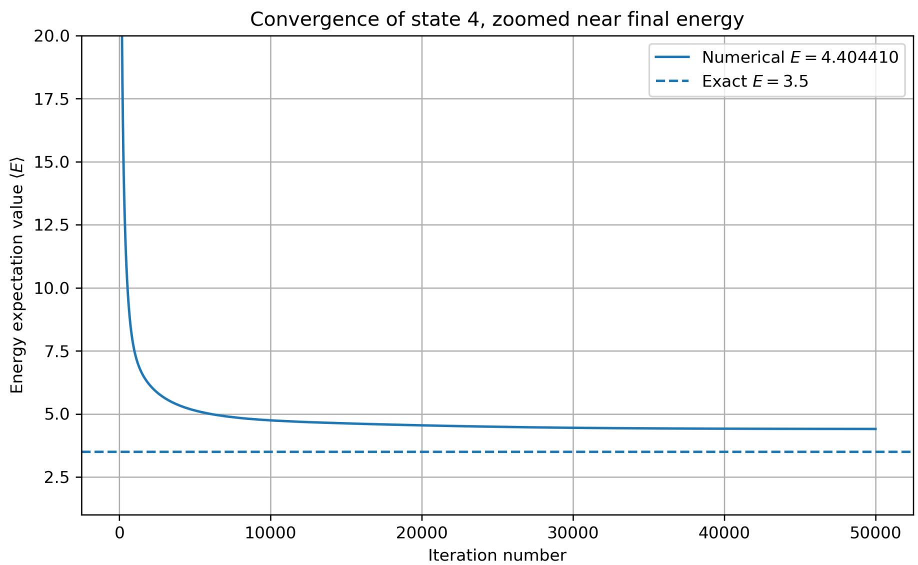

Figure 8: Energy convergence of the fourth state.

2.3 Wavefunction visualizations

The spatial structures of the calculated wavefunctions are shown in Figures 9–12. The first wavefunction corresponds to the ground state and has no internal nodes. The second state has one nodal line in the x direction, as expected for the state $(n_x, n_y) = (1, 0)$.

The third and fourth states correspond to the degenerate energy level

$$E = 3.5.$$

This level may correspond to either

$$(n_x, n_y) = (2, 0)$$

or

$$(n_x, n_y) = (0, 1).$$

Because these states are degenerate, the numerical method may produce any orthogonal linear combination of them. Therefore, the exact visual form of the third and fourth wavefunctions may depend on the random initial condition and the orthogonalization process.

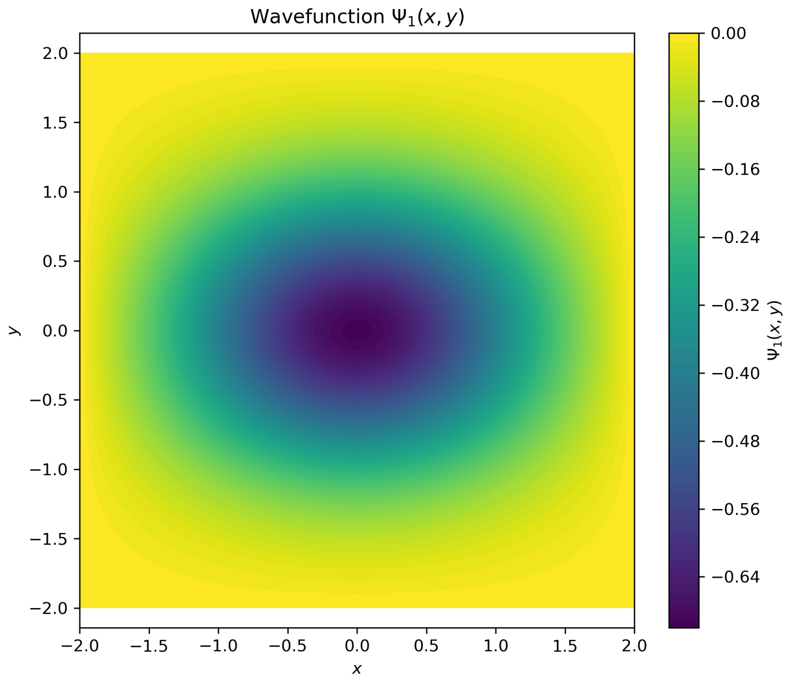

Figure 9: Wavefunction of the first state.

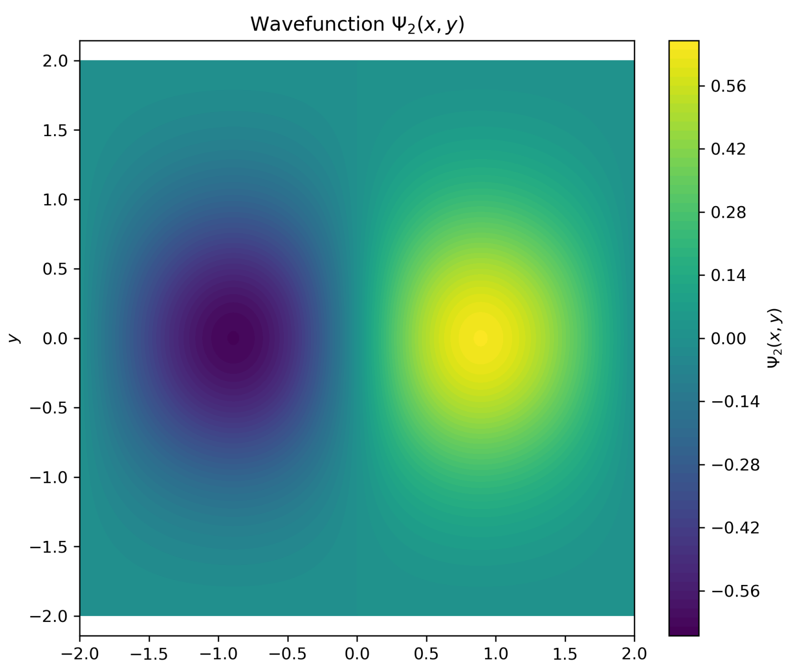

Figure 10: Wavefunction of the second state.

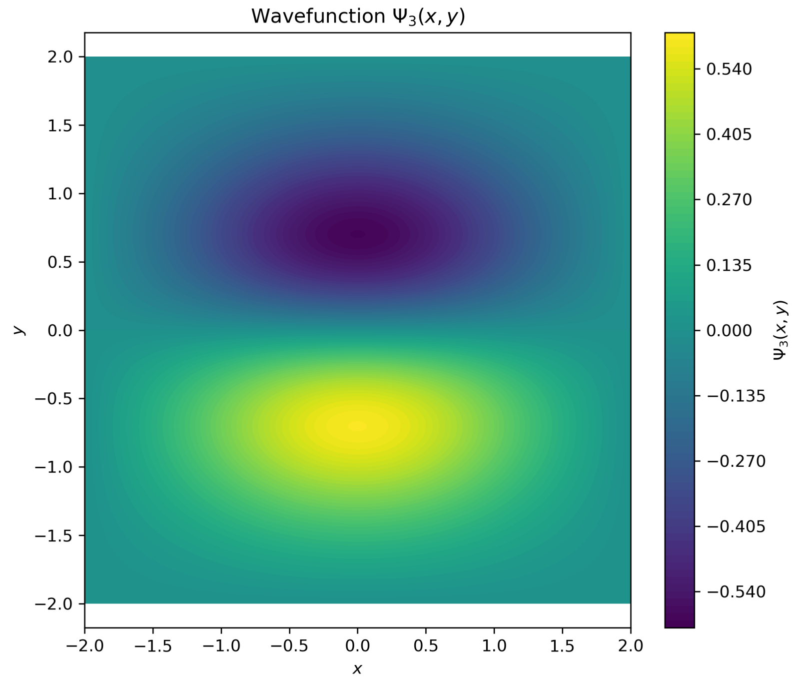

Figure 11: Wavefunction of the third state.

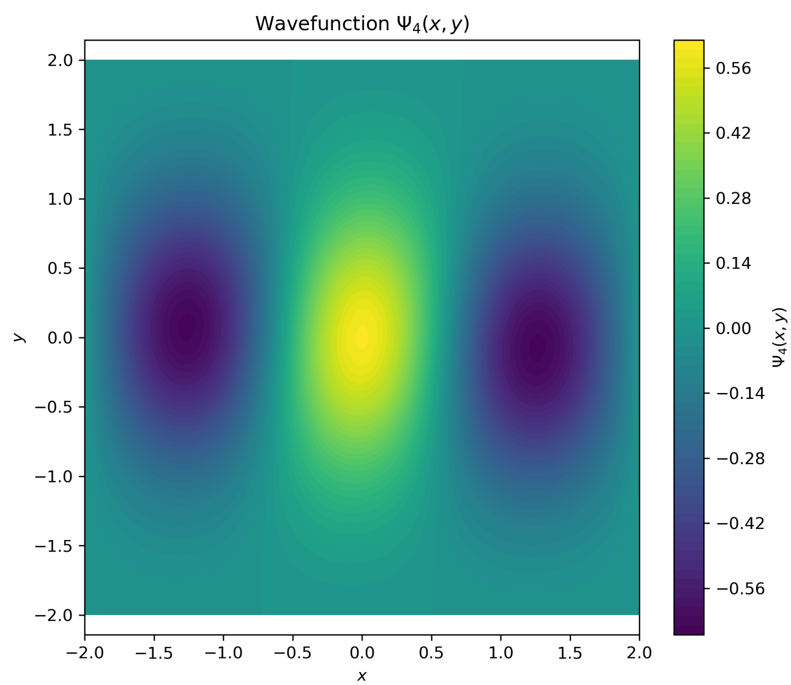

Figure 12: Wavefunction of the fourth state.

2.4 Comparison with exact energies

The final numerical energies were compared with the exact analytical spectrum of the two-dimensional anisotropic harmonic oscillator. The results are summarized in Table 1.

State	Expected dominant state	Exact energy	Numerical energy
1	$(0, 0)$	1.500000	1.539390
2	$(1, 0)$	2.500000	2.766640
3	$(2, 0)$ or $(0, 1)$	3.500000	3.564180
4	$(0, 1)$ or $(2, 0)$	3.500000	4.404410

Table 1: Comparison between exact and numerical energies for the four lowest calculated states.

The first state gives a numerical energy of 1.539390, which is close to the exact ground-state energy $E_{0,0} = 1.5$. The absolute error is

$$|E_{\text{num}} - E_{\text{exact}}| = 3.9390 \times 10^{-2}.$$

This confirms that the method correctly converges toward the ground state.

For the second state, the numerical energy is 2.766640, while the exact value is 2.5. The error is larger,

$$|E_{\text{num}} - E_{\text{exact}}| = 2.6664 \times 10^{-1}.$$

This state is more sensitive to numerical errors because it contains a nodal structure and requires orthogonalization with respect to the ground state.

The third calculated state gives $E_{\text{num}} = 3.564180$, which is close to the exact degenerate energy level $E = 3.5$. The corresponding error is

$$|E_{\text{num}} - E_{\text{exact}}| = 6.4180 \times 10^{-2}.$$

This indicates good convergence to one of the states belonging to the degenerate level.

The fourth state has the largest deviation from the exact value. Its numerical energy is 4.404410, while the expected energy is 3.5, giving

$$|E_{\text{num}} - E_{\text{exact}}| = 9.0441 \times 10^{-1}.$$

This suggests that the fourth state did not converge as accurately as the lower states. A likely reason is the degeneracy of the third and fourth exact states, which makes the orthogonalization procedure more sensitive to numerical errors. The result may also be affected by finite grid spacing, finite box size, and incomplete convergence.

Overall, the first and third states show good agreement with the analytical values, while the second and especially the fourth state have larger numerical errors. The results still demonstrate the main behavior of the imaginary-time method: lower-energy states are obtained reliably, while higher and degenerate states require more careful numerical treatment.

3 Conclusions

The imaginary time method was successfully used to obtain the lowest energy states of the two-dimensional anisotropic harmonic oscillator. The ground-state calculation showed that the parameter α strongly affects convergence: small values give slow but stable convergence, while values close to α_c converge faster but may become unstable if chosen too large.

The computed ground-state energy was close to the exact analytical value. Excited states were obtained by adding orthogonalization to the iteration procedure. The results showed reasonable agreement with the expected spectrum, especially for the lower states. Larger deviations for higher states are mainly caused by finite grid effects, finite box size, degeneracy, and the sensitivity of the orthogonalization procedure.